

DILAN JAYESHBHAI PATEL

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Education

The University of Manchester

3rd Year, On track for 1st Class

Sept. 2024 - Present

- **Relevant Modules:** Advanced Distributed Systems, User Experience, Graphics & Virtual Environments, Agile Software Pipelines, Cognitive Robotics, Computer Vision
- **Final Year Project:** Comparative analysis of human pose estimation models for exercise repetition counting
 - * Developed an **LSTM-based model** to classify exercise phases and count repetitions, achieving **92% accuracy**
 - * Curated a dataset of **75+ workout videos** from diverse sources and implemented a structured feature extraction pipeline
 - * Optimized model performance through hyperparameter tuning and leveraged **PyTorch** and **TensorFlow** for deep learning

The University of Manchester

2nd Year, Achieved 2:1

Sept. 2022 - June. 2023

- **Relevant Modules:** Software Engineering 1 & 2, Machine Learning, Algorithms & Data Structures, Database Systems, Intro to Visual Computing

1st Year, Achieved 2:1, with 87.6% in my First Year Project

Sept. 2021 - June. 2022

- **Relevant Modules:** Mathematical Techniques, Fundamentals of Computer Engineering, Data Science, Intro to Programming 1 & 2
- Designed a website game **Guess or Drink** using **HTML, CSS, JavaScript & Socket.IO**
 - * Collaborated with 6 students over the course of a year, presenting our work throughout
 - * Individually added new features to the website during Summer 2022

The City of Leicester College

*A-Levels, Achieved A*A*AA with A* in Computer Science*

Sept. 2019 - June. 2021

- **Subjects:** Computer Science, Further Mathematics, Mathematics, Physics
- Built a 3D Aim Trainer game using **Unity** and **C#** in a 6-month time frame

GCSE, Achieved grades 8-4 in 10 subjects, including English, Mathematics, and Computer Science

Sept. 2014 - June. 2019

Experience

Atmos International

June. 2023 - June. 2024

Software Engineer Placement

- Designed an interactive schematic component in **Blazor** (using **C#** and **F#**) showcasing **full** and **collapsed** views of pipelines
- Successfully integrated this into an **upcoming project**, with plans to maximise its **visual potential** when working with **clients**, providing a clearer understanding of their pipelines and potential issues
- Playback slider which moves through time and **displays metrics** (e.g. Pressure, Temperature, Standard Flow) for instruments within a pipeline
- In-depth experience with the **.NET framework**, utilizing **Azure DevOps**, **Git workflows**, and actively contributing to **code reviews** and **pull requests**

ADVNC

Summer of 2022

Virtual Assistant

- Improved social media publicity by **10%**, interacting with existing and potential clients
- Performed quality assurance for websites and blogs, devising fitting solutions to problems
- Organised a clients business and personal expenses for **3 months**

Nuffield Research Placement

Summer of 2020

Twitter Data Analysis

- Used **Python & Google Colab** to fetch and analyse tweets from Twitter using **REST API Tweepy**
- Applied sentiment analysis to evaluate tweets, presenting insightful data through graphical representations using **Matplotlib.pyplot** and **TextBlob**
- Shared my knowledge through **blog publications**, attracting **3200 viewers**

Personal

- Currently in the process of completing the **AWS Cloud Practitioner** course using **AWS Skill Builder**
- I love to complete programming courses on **Udemy** with programming languages like **Python** and **Java**
- Enjoy listening to podcasts about Physics and Artificial Intelligence as I am able to learn about the fundamentals to our universe whilst also feeling inspired by the great potential we have to develop our future using AI